Thin QR and full QR

Both reconstruct A exactly. The full form keeps $m - n$ extra columns of Q that multiply against rows of zeros — useless for the product, and the only route to a basis for the left null space.

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FORMS

TYPE	CONDITION	Q	R	BASES OBTAINED	USED FOR
THE WORKING FORM					
 **Thin QR**	$m \geq n$, columns of A independent	$m \times n$, orthonormal columns	$n \times n$ upper triangular	a basis for $\mathrm{Col}(A)$	solving, lea squares
THE COMPLETE FORM					
 **Full QR**	same A, Q completed to a square	$m \times m$, fully orthogonal	$m \times n$ — triangular block, then zeros	bases for $\mathrm{Col}(A)$ and its complement	left null spa theory

The extra columns are not redundant so much as unused. They complete Q into a genuine orthogonal matrix, which is what makes $QQ^{\mathsf{T}} = I$ hold as well as $Q^{\mathsf{T}}Q = I$ — and that second identity is what any argument about the left null space needs.